Training Generative AI LLMs

- Author: Lijing Cui
- LLM training lifecycle
- Resources and costs
- Major AI models
- Data to deployment
- Industry impact

What Are LLMs?

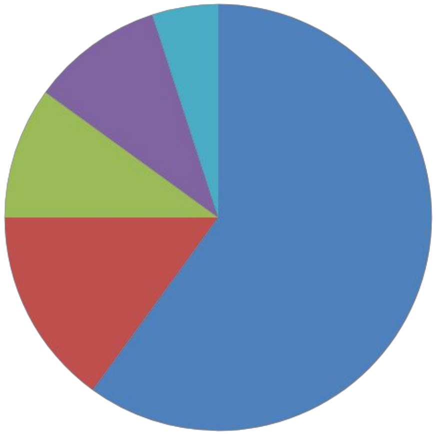
- Generate human-like text
- Learn from massive datasets
- Support coding tasks
- Answer questions
- Translate languages
- Power AI assistants

Training Process

- Collect raw data
- Clean and filter data
- Tokenize text inputs
- Train neural network
- Fine-tune performance
- Deploy to users

Data Sources

- Web pages
- Books and journals
- Source code
- Research papers
- Public datasets
- Licensed content



Tokenization

- Break text into tokens
- Convert words to IDs
- Reduce complexity
- Enable model learning
- Improve efficiency
- Support predictions

Computing Resources

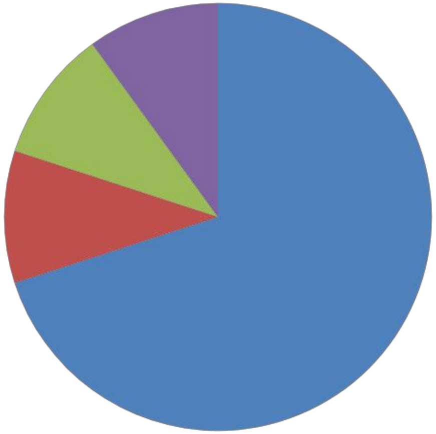
- Thousands of GPUs
- High-performance clusters
- Large storage systems
- Fast networking
- Cloud infrastructure
- Massive parallel processing

Energy and Time

- Training lasts months
- Millions of kWh used
- Continuous computation
- Cooling requirements
- High operational cost
- Environmental impact

Training Costs

- GPT-4 costs millions
- Hardware dominates spending
- Data preparation costs
- Research staff expenses
- Infrastructure maintenance
- Ongoing optimization



Deployment

- Serve through APIs
- Power chatbots
- Support applications
- Monitor performance
- Update models
- Scale globally

Conclusion

- Data drives quality
- Computing drives learning
- Costs remain significant
- Tuning improves accuracy
- Deployment creates value
- Future growth continues